



Hex ukulele



Tomek

[VIEW IN BROWSER](#)

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Summary

Soprano ukulele - No supports, no glue, no hardware required (except strings). Fits on standard 3d printer.



23.11 hrs



12 pcs



0.20 mm
0.15 mm
0.25 mm



0.60 mm



PLA
PET



460 g



Anycubic
Mega S

[Hobby & Makers](#) > [Music](#)

Tags: [hexagon](#) [ukulele](#) [pickup](#) [piezo](#) [musicalinstrument](#)
[pegs](#) [soprano](#) [prusalele](#)

My previous design - [heart shaped ukulele](#) is working great and i love it's sound. So i decided to make another ukulele project: ukulele shaped ukulele, inspired by that design. No more hearts. This design is an attempt to join some classical ukulele forms with some hexagon details.

The goals are pretty much the same:

- Can be 3d printed on **standard printer** (this time all parts fits on 215x215 bed or 210x240)

- **No additional parts** (Nothing except string is required - no tuning pegs, bridges, bolts, nuts etc., **everything is 3d printed** (you can add carbon fiber strips but it is optional))
- **No glue** - all parts can be assembled and disassembled with 3d printed bolts and joints. This is great for future improvements and upgrades. You can always use some glue for rigidity, but it is not necessary.
- **No supports needed** - all parts are designed to be printed without supports (just print parts, assemble put the strings on, tune it and play).
- **Nice shape** - No more hearts - just classical shape ukulele and some hexagon style
- **Full feature** 4-string ukulele - scale length is 350mm

This one is a bit more challenging. You have **multicolor** printing (on single nozzle printer), weird **overhangs** and a bit of **bridging**. But don't worry. Everything is designed to be printed as easy as possible.

Does it work? You can hear and see all three of my 3d printed ukuleles in action on the video below:

BOM:

- about 400g of PLA
- about 50 of PETG (can also be PLA)
- one pack of soprano ukulele strings (I love Aquila Red Series - both Low and High G)
- (optionally) 2 pieces of carbon fiber strips 2x8mm (about 37 cm length each)

General

All parts can be printed **without support**. All parts can be printed with **0.4 or 0.6 nozzle**. All parts can be **printed on a regular 220x220** printer. I have printed mine with PLA (except internal-rod and other small threaded parts - which for better layer adhesion I've printed with PETG). PETG can be too soft for neck and body, but you can give it a try - especially if you plan to use carbon fiber strips. For pegs and dots (small screw) you can add brim to ensure it stays on the bed.

Orientation

Orient all the parts with largest flat side on the bed (all files should be already oriented for printing).

Ukulele version 1.2 info

I started improving my design using feedback from users. It's generally same ukulele but with some small improvements and tweaks. I left files from 1st version intact just in case someone need them or want to compare which version is better. The new ukulele should look same (except sound holes), sound the same (quality is still great, but my electroacoustic version is a bit quieter. I'm still not sure why) - but everything is just slightly adjusted. Tolerances are better, internal construction is reduced. Also there is now possibility to print electroacoustic version. All improvements are mentioned in the part list below. For the electroacoustic version you have to get piezo preamp/tuner set similar to this:

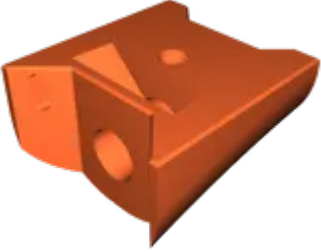


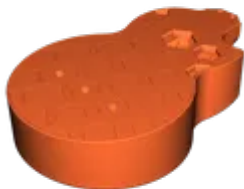
Printing instruction and tips:

General information

If your printer and filament is not calibrated well there can be problem with threaded parts that needs to be printed precisely. But do not fear. Just print big parts (neck, neck-middle and body) and then small ones (dots, internal-rod, gears and pegs. If you're lucky - everything will fit together. If not - just reprint those smaller parts with XY Size compensation (+/- 0.1 or less should be enough). As a result the internal-rod should fit pretty tight. dots should fit with small friction and gears and pegs - with little or no friction (but still tight). If You print pegs/gears with PLA - consider adding some grease or PTFE Lube - to make sure they don't fuse from the friction.

Start with bigger parts:

	neck-middle.stl start with this piece to check if everything is ok and carbon strips fits (if you plan to use them). Version 1.2 contains several versions of this part - the ends of this part are skewed by a fraction of degree to compensate string tension. Try 0.2 or 0.3 at the beginning - if you find that the compensation is too strong or too weak - reprint it . probably you don't need any compensation when using carbon strips. Also the slot for carbon strips is slightly bigger
	neck.stl just print it flat on the bed. You can use variable layer height to make the surface smoother. Version 1.2 has corrected peg holes placement so gears are aligned with neck face and stay in place better. Slot for carbon strips is slightly bigger. threaded holes for dots ale 1mm longer
	body.stl



Multicolor for single nozzle/extruder:

For **multicolor body**: Open stl in Prusa slicer. Split to objects, Merge all the hexagon parts into one object (There is a [video](#) in my [hex drum](#) project showing how to do this operation). Set z-hop to 0.4mm or more. Print hexagon parts first and then print rest of the body (turn off supports). You can also join two gcode files with M600 command between to ensure exactly same homing position (you have to remove some parts of g-code from both files).

WARNING!!!: If You have mesh bed leveling at the beginning of every print You have to figure out how to turn this off for the second print (merging gcode files method should be ok if you remove homing procedure from second print). In any case please stay and watch the print during color change to make sure there are no serious collisions.

It is also good idea to do research on the internet. There are several methods to achieve same result. You have to figure out what is best for your 3d Printer. Here are some links to start with:

- <https://www.youtube.com/watch?v=9F2uaDW0W4c>
- <https://www.youtube.com/watch?v=PQRaSvQdVFY> - this one works great for my new MK4
- <https://www.youtube.com/watch?v=D7FiT5VqLSg>

You can also print it with multicolor 3d printer or just **as single color part**.

3d printers with dual extrusion/ multicolor support:

You can for sure use those features of your 3d printer. However as I don't have

	any experience with such a printers I cannot give you any clues, but there is a great community around this project. Just check if someone with 3d printer like yours have made a successful build. If so you can always ask for some tips. Other hints: If you want to preserve nice back curvature you can use Variable Layer Height - but remember to define Minimum Shell Thickness in mm (even 5 top layers is pretty thin when layer height is 0.07mm. Version 1.2 has (apart from slightly bigger carbon strip slots) less construction elements inside. It comes with two flavours acoustic and electroacoustic (body_e.stl) and because of this, sound holes had to be rearranged. If you want ukulele that looks exactly like on the pictures use previous version. For Cura users there are 3mf files in Stuff folder.
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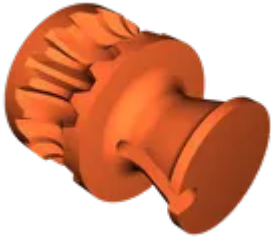
Now it's time to print smaller threaded parts:

	internal-rod.stl
	I recommend to print this in PETG. Two PLA parts can fuse together on threads due to friction and low melting temperature. Version 1.2 has just some tiny adjustments with tolerances.

	dot7.stl, dot.stl You need 6 of these, but they are fragile, so it's nice to have a couple of spares. These are also better when printed in PETG. Version 1.2 has improved tolerances and extended thread length to 8mm.
	bridge_bolt.stl You need 3 of these. You can print four and use one as a tap to clean threads in Body part. These are also better when printed in PETG. Version 1.2 has looser tolerances.

Tuning system:

	peg.stl, pegx4.stl You can print 4 pieces of single peg.stl, but I recommend printing pegx4.stl, because the connections between pegs reduce the chance of failure (these are pretty easy to be knocked off front bed by the print head). I think it's better when printed with PETG, but I've printed mine with PLA and they work great. Remember that pegs stay in place only because of string tension so before winding they are loose. Version 1.2 has tighter tolerances (pegs are a bit thicker). Some users have problems with printing those pegs. For them there are pegs designed for easy print with larger bed contact surface. You can find them in "Stuff" folder in failureproof-peg.stl file.
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	gear_G.stl gear_C.stl gear_E.stl gear_A.stl gear_blank.stl You need 4 Gears. If you plan to use standard GCEA tuning you can use version with string names (add color change at 19.5mm for different color of letters). Version 1.2 has tighter tolerances, redesigned shape for better grip and polished letters. This is a bit taller so color change is at 20.0mm. I've also added blank gear
	gear_screw.stl This is a screw that secures the string at the bottom of the gear - so you need four of them. This also is best to be printed with PETG for good layer adhesion. Version 1.2 has improved tolerances.

An most important parts for sound:

	fretboard.stl Print it flat on bed. Add color change at 3.5mm for different color of frets. Version 1.2 has improved grooves in nut and slightly shorter frets. The slots in the nuts are pretty thin. If your strings don't fall deep into grooves - use file or knife to widen the gaps. Also check "zero fret" version in "Stuff" folder - this keeps tuning better for me.
	bridge.stl Print it flat on the bed. Version 1.2 has significantly higher bridge.

	saddle.stl
	there are two versions - for lowG and standard tuning. The idea is to compensate string thickness (like many saddles do). Print it wyth some hard material because strings will put some force on it. Version 1.2 has redesigned shape that should be easier to print

Optional

	tensioning-strips.stl
	If you don't plan to use real carbon fiber strips i highly recommend using these. Print this with PLA or other stiff material. Use it as a replacement of carbon strips. When putting ukulele together place those strips bended backwards so the strips tension works against string tension.

Assembly steps:

Here two videos of how to assembly the ukulele. First is the assembly of the electroacoustic hex ukulele. This was released recently - and all the users so far used the previous video of heart ukulele assembly, which can explain certain things better. the assembly of both ukulele is pretty much the same, but **internal-rod** is not ideally symmetrical - so do three turns only in body part - and after that attach neck. Also now strings are secured with screws. Text description below the videos.

1. Take **neck**. Insert two strips (printed, carbon fiber or no strips at all)
2. Take **body** and insert **internal rod** and screw in for two or three turns
3. add **neck-middle** and **neck** with optional strips
4. Using small allen key start turning **internal rod** through the hole in **neck-middle** part
5. When connection is tight add **fretboard** and screw it with **dots** (use small flat screwdriver - do not overtight).
6. Use **bridge-bolts** to connect **bridge** to the **body**.
7. Insert the **saddle** to the bridge (should fit tight)
8. time to add the strings - make knot on one end of the string (I use the Figure 8. knot)
9. Put this end in proper gap on the string

10. lead string through the hole in head and then through the **gear** (thinner end first)
11. Secure the end of the string with the **gear-screw**
12. place the gear in proper hole in the head
13. secure the gear with a peg and start turning the peg (keep the string tensioned)
14. after a couple of gear turns, the string should be tensioned
15. Repeat these steps for three other strings
16. That's all - You have a 3D printed ukulele.

Keep in mind that the strings **will get out of tune** quickly at the beginning. Be patient and tune your ukulele many times and eventually it will stay in tune for longer.

Here is a video showing how to attach the string with this version of the tuning mechanism:



Updates:

24.07.2023:

Uploaded assembly video

21.07.2023:

Uploaded source files (step and fusion 360)

08.07.2023:

The project won first prize in Printables contest for musical instruments.

01.06.2023:

New version with many changes:

1. additional electroacoustic version for set similar to this:
 - <https://www.aliexpress.com/item/1005004641085743.html>
2. better tolerances for dots, bolts, gears and pegs
3. middle part with some angles to compensate string tension
4. slightly wider string gaps in nut
5. reranged soundholes (for compatibility with tuner)
6. 1mm longer dots
7. gears are now properly aligned with head
8. redesigned bridge with optional space for piezo pickup

28.04.2023:

I added small spacers. If your neck bends:

1. remove strings and unscrew fretboard
2. try once more tightening the rod
3. bend the neck backwards and slide the spacer in the gap (smaller spacer goes between neck and neck-middle)

This should keep the neck more straight

I've added experimental tensioning strips:

You can use it instead of carbon strips.. The idea is to create backward tension that will be compensated by string's force.

Final thoughts

I think the the project is a success. You get fully printable soprano ukulele. You can modify any part, disassemble and assemble it. There is still a room for improvements so all remixes and upgrades are welcome.

Need help?

If You have any questions - don't hesitate to ask. I always try to answer questions and help.



Model files



Visualization

1 file



ukulele-all-elements.3mf



Sources

3 files

hex_ukulele_acoustic_v12.step

hex_ukulele_electroacoustic_v12.step

ukulele.f3z



Ukulele Parts v 1.2

26 files



neck-middle-00.stl



neck-middle-01.stl



neck-middle-02.stl



neck-middle-03.stl



neck-middle-04.stl



neck-middle-05.stl



neck-middle-06.stl



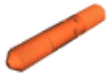
neck.stl



body.stl



body_e.stl



inner-rod.stl



dot.stl



bridge-bolt.stl



peg-x4.stl



peg.stl



gear-g.stl



gear-c.stl



gear-e.stl



gear-a.stl



gear-blank.stl



gearscrew.stl



fretboard.stl



bridge.stl



bridge_e.stl



saddle-low-g.stl



saddle-high-g.stl



Ukulele Parts

17 files



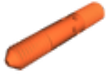
body.stl



neck.stl



neck-middle.stl



inner-rod.stl



peg-x4.stl

☐ Easier to print than 4 separate objects



bridge.stl



fretboard.stl



peg.stl



dot7.stl

☐ Print 6 pieces



gear-g.stl



gear-c.stl



gear-a.stl



gear-e.stl



gear-screw.stl



bridge-bolt.stl

☐ Print 3 pieces



saddle-high-g.stl



saddle-low-g.stl



Stuff

9 files



body-with-splited-objects.3mf



spacers.stl



tensioning-strips.stl



bridge-bolt-loose-tolerance.stl



failureproof-peg.stl



box.3mf

☐ Body with splitted objects. Ukulele 1.2 RC1



box_e.3mf

☐ Body with splitted objects. Ukulele 1.2 RC1 - Electroacoustic



fretboard-fretless.stl

☐ experimental fretboard for adding fretwire or to be used as fingerboard for fretless



fretboard-zero-fret.stl

Print files



Print files - 0.6 nozzle

12 files



bridgenut.gcode

PLA 0.60 mm 0.20 mm 0.43 hrs 4 g



pegsx4.gcode

🌀 PLA 📏 0.60 mm 📐 0.20 mm ⌚ 1.53 hrs ⚖️ 10 g



7dots.gcode

🌀 PLA 📏 0.60 mm 📐 0.20 mm ⌚ 0.51 hrs ⚖️ 8 g



pegbolts.gcode

🌀 PLA 📏 0.60 mm 📐 0.15 mm ⌚ 0.42 hrs ⚖️ 4 g



gears.gcode

🌀 PLA 📏 0.60 mm 📐 0.25 mm ⌚ 0.82 hrs ⚖️ 9 g



fretboard.gcode

🌀 PLA 📏 0.60 mm 📐 0.20 mm ⌚ 1.39 hrs ⚖️ 35 g



neck-middle.gcode

🌀 PLA 📏 0.60 mm 📐 0.20 mm ⌚ 1.22 hrs ⚖️ 24 g



rod.gcode

🌀 PET 📏 0.60 mm 📐 0.20 mm ⌚ 0.40 hrs ⚖️ 5 g



neck.gcode

🌀 PLA 📏 0.60 mm 📐 0.15 mm ⌚ 4.61 hrs ⚖️ 95 g



ukulele_mix.gcode

🌀 PLA 📏 0.60 mm 📐 0.20 mm ⌚ 12.00 hrs ⚖️ 266 g

📄 this is mix of hexagons and main body part

ukulele-light.gcode



🌀 PLA 📏 0.60 mm 📏 0.20 mm ⌚ 11.45 hrs ⚖️ 259 g

ukulele-dark.gcode



🌀 PLA 📏 0.60 mm 📏 0.20 mm ⌚ 0.33 hrs ⚖️ 7 g

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